

Operation Phantom Circuit – CTF Solver Guide

This guide contains methodology, commands, and script templates for Tasks 1–4. It is a workflow guide; actual flags require the challenge artefacts.

Task 1 – Ghost Signal

```
binwalk ghost_signal_A.bin
strings ghost_signal_A.bin > A.txt
strings ghost_signal_B.bin > B.txt
strings ghost_signal_C.bin > C.txt

# Compare sections
cmp ghost_signal_A.bin ghost_signal_B.bin
cmp ghost_signal_B.bin ghost_signal_C.bin

# Extract RSA parameters and investigate broadcast attack
grep -Ei "rsa|modulus|public|cipher" A.txt
```

Likely approach: identify common plaintext encrypted under different RSA moduli and test CRT/Hastad broadcast attack.

Task 2 – Ghost Relay

```
binwalk relay_fw_v3.bin  
strings relay_fw_v3.bin > relay.txt  
xxd intercept_v3.bin
```

```
# Search secrets  
grep -Ei "serial|device|seed|secret" relay.txt
```

Recover protected configuration region, identify device secret, derive session key, reconstruct nonce, decrypt packet.

Task 3 – Firmware Recon

```
file init_v2
readelf -h init_v2
rabin2 -I init_v2

# Reverse transformation
ghidra init_v2

# Decode filesystem
unsquashfs decoded.squashfs

# Recover credentials
cat squashfs-root/etc/shadow
john hash.txt --wordlist=/usr/share/wordlists/rockyou.txt
```

Trace boot chain, recover Wi-Fi credentials, identify security issues, and assemble composite flag.

Task 4 – Orpheus Hardened

```
binwalk -Me orpheus_fw_v5.bin
```

```
tshark -r relay_traffic_v4.pcap -T fields -e data > payloads.txt
```

```
# Recover hidden binaries and reverse in Ghidra  
ghidra recovered_binary
```

Analyse payload structure, derive session material, compute flag offset, and decrypt FLAG{...}.